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Examining Files and Finding Hidden Data



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UTM BIG DATA CENTRE
TOGETHER WE MAKE MOTIONS...

Menginovasi Penyelesaian Lestari

DFBOOK DIGITAL FORENSICS · LESSON 5.0

Examining Files and Finding Hidden Data

Lesson overview: from stored data to findings and a timeline



LESSON 5.0 · OVERVIEW

Examining Files

Turning stored data into findings and a first timeline

BY THE END YOU WILL BE ABLE TO

Identify a file by its header

And read the metadata inside a photo

Search and carve

Find keywords, and recover files from free space

Describe how data is hidden

And how each technique is detected

Filter known files and build a timeline

Then write an examination plan

LESSON 5.0 · ROUTE

Four Sub-Lessons

Three hands-on labs and one demo

5.1	**File Signatures and Photo Metadata**	Lab: a renamed JPEG and its Exif data
5.2	**Keyword Searching and Carving**	Lab: search and carve the image
5.3	**How Data Is Hidden**	Demo: hidden text and a stego image
5.4	**Filtering Known Files and Building a Timeline**	Lab: hash filter and six timestamps

Labs use the training image only. Work on the verified copy; stop and report if you find illegal material.

LESSON 5.0 · THE CASE

The Teratai Case So Far

From recovered files to findings

SO FAR

The examiner found the partitions, recovered a deleted invoice, and read the Recycle Bin and web history.

NOW

The examiner must find renamed and hidden files, search the whole image, and put events in order.

STILL TO DO

- **Find files with false extensions**
- **Search for the supplier's name**
- **Check for hidden data**
- **Build a first timeline**

The question for this lesson: what do the files really contain, and in what order did things happen?

UP NEXT

File Signatures and Photo Metadata

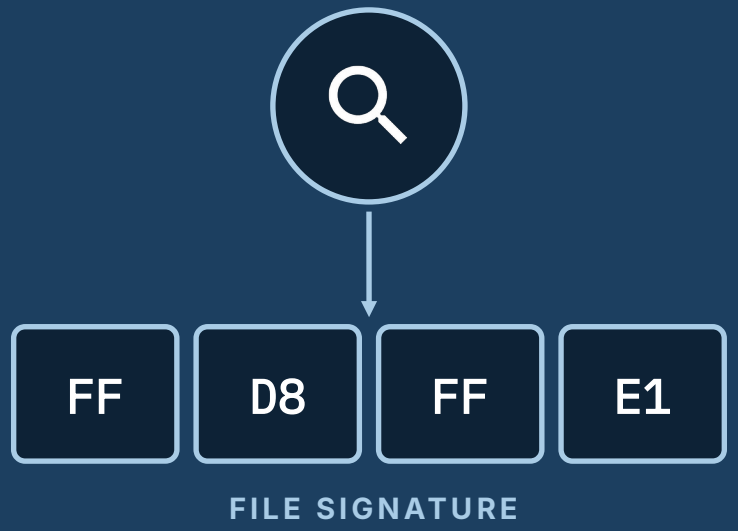
We start with how to tell what a file really is, and what a photo records about itself.

- **Hexadecimal and file headers**
- **Renamed files**
- **Photo metadata (Exif)**

DFBOOK DIGITAL FORENSICS · LESSON 5.1

File Signatures and Photo Metadata

What a file really is, and what a photo says about itself



IN THIS LESSON

1. Hexadecimal and file headers
2. Renamed files
3. Photo metadata (Exif)

LESSON 5.1 · OVERVIEW

File Signatures and Metadata

A file's name can lie; its first bytes rarely do. This lab shows how to identify a file from its contents, and what a digital photo records about itself.

BY THE END YOU WILL BE ABLE TO

1. Read file headers in hexadecimal
2. Identify a renamed file by its header
3. Read and judge a photo's Exif metadata

INTRODUCTION · WHY THIS MATTERS

Why This Matters

Illustrative: a document that will not open

`site_visit.docx`

Word says it cannot open it.

→

ITS FIRST BYTES, IN A HEX EDITOR

FF	D8	FF	E1	...

FF D8 is the start of every JPEG image. The file is a photo with a false extension.

Renaming a file's extension is one of the oldest ways to hide it, and one of the easiest to catch.

Trust the contents of a file, not its name.

INTRODUCTION · ROADMAP

Your Route Through Lesson 5.1

Each part answers one question, in order

1

PART 1

Hexadecimal and file headers

How is a file's type written in its first bytes?

2

PART 2

Renamed files

How do you catch a file with a false extension?

3

PART 3

Photo metadata (Exif)

What does a photo record, and can you trust it?

One case runs through the module: Teratai Supplies, where auditors have found payments for materials that were never delivered.

PART 1 OF 3

Hexadecimal and File Headers

Reading the first bytes of a file

01

IN THIS PART YOU WILL LEARN TO

- Read bytes in hexadecimal
- Recognise common file headers

KEY TERMS

Hexadecimal **Byte** **File header**

PART 1 · CONCEPT

Hexadecimal: A Quick Recap

Base 16: a compact way to write the value of each byte

Digits	0 to 9, then A, B, C, D, E and F for ten to fifteen
One byte	8 bits, written as exactly two hexadecimal digits
Smallest and largest	00 is 0; FF is 255 (binary 11111111)
Written as	0x45, or simply 45 in a hex editor
Offsets	A byte's position in a file, counted from 0

Windows Calculator in Programmer mode converts between decimal, hexadecimal and binary.

PART 1 · CATEGORIES

Common File Headers

The first bytes that identify a file's real type

File type	First bytes (hex)	Seen as text
JPEG image	FF D8 FF	(not readable text)
PNG image	89 50 4E 47	.PNG
PDF document	25 50 44 46	%PDF
Office file or ZIP	50 4B 03 04	PK (DOCX and XLSX are ZIP files)

File header: the first bytes of a file, which identify its format. Also called a file signature.

PART 1 · STRUCTURE

Inside a JPEG File

Fixed markers at the start and the end

FF	D8	FF	E1	. .	. .	45	78	69	66
START OF IMAGE		TYPE MARKER				LABEL: E x i f			

Offset 0	FF D8: every JPEG starts here
Offset 2	FF E0 for a standard JPEG (JFIF); FF E1 for an Exif JPEG
Offset 6	The label name: JFIF or Exif
The end	FF D9: the end-of-image marker

These markers also let tools find and carve whole images from free space, as Lesson 5.2 shows.

PART 2 OF 3

Renamed Files

When the name and the contents disagree

02

IN THIS PART YOU WILL LEARN TO

- Explain why extensions can mislead
- Spot a header that does not match

KEY TERMS

Extension **Signature mismatch**

PART 2 · CONTRAST

Extension and Header

One can be changed in a second; the other describes the contents

THE EXTENSION

Part of the name

- Tells Windows which program to open
- Anyone can rename it
- A spreadsheet renamed .jpg will not open

THE HEADER

Part of the contents

- Written by the program that made the file
- Unchanged by renaming
- Read in a hex editor or forensic tool

Forensic tools compare the two and flag any mismatch. Autopsy, for example, has an extension mismatch detector.

PART 2 · WORKED EXAMPLE

Three Files, Three Headers

Illustrative: do the names match the contents?

File name	First bytes	Verdict
`minutes.docx`	`50 4B 03 04`	**Matches: DOCX files are ZIP files**
`budget.jpg`	`25 50 44 46`	**Mismatch: it is really a PDF**
`notes.txt`	`FF D8 FF E1`	**Mismatch: it is really an Exif JPEG**



ANSWER

Two files are disguised. Record both names and headers, then open each with the right viewer.

PART 3 OF 3

Photo Metadata (Exif)

What a photo records about itself, and how far to trust it

03

IN THIS PART YOU WILL LEARN TO

- List what Exif metadata can hold
- Judge whether a photo's time is reliable

KEY TERMS

Exif **Metadata** **UTC**

PART 3 · CATEGORIES

What Exif Metadata Holds

Details a camera or phone writes into each photo

The device

Make, model, sometimes serial number

The settings

Shutter speed, focal length, resolution

Date and time

When the photo was taken, by the device clock

Location

Latitude and longitude, if GPS was on

Exif: the Exchangeable Image File format, a standard for storing metadata inside JPEG and TIF photos. Most devices save photos as Exif JPEGs.

View it with Autopsy, ExifTool or another Exif viewer. High-end cameras may also save raw files, which need the maker's software.

PART 3 · CONCEPT

How Far to Trust a Photo's Time

Exif dates come from the device's clock, which can be wrong

The clock was set wrongly

By mistake, or on purpose to mislead.

The clock reset

After a battery or electronic failure.

A different time zone

Some devices record Coordinated Universal Time (UTC).

The file was converted

Copying in another format can lose or change metadata.

Look for corroboration: other records of the same event, the light and shadows, or the place the photo shows.

PART 3 · WORKED EXAMPLE

A Photo Timed in UTC

Illustrative: a delivery-yard photo whose device records UTC

THE EXIF DATA

Date and time

`2026:03:03 02:30:00`

Device clock

`Set to UTC`

Location

`Johor, Malaysia`

1. **Find the local offset**
   Malaysia is UTC+8 all year
2. **Convert the time**
   02:30 + 8 hours = 10:30 local time
3. **Check it makes sense**
   Daylight in the photo fits mid-morning

ANSWER

10:30 local time on 3 March. Report both values, and say how you converted them.

PART 3 · LAB

Finding the Renamed Photo

Case thread: a renamed JPEG on the Teratai training image

1. **Recap hex**
   Convert a few bytes between hex, decimal and binary.
2. **List mismatches**
   Run Autopsy's extension mismatch check on the image.
3. **Check the header**
   In the Hex view, look for FF D8 at offset 0, and FF D9 at the end.
4. **Export a copy**
   Save the file to your export folder with a .jpg extension.
5. **Read its Exif**
   Note the device, date, time and any location.
6. **Record it all**
   Original name, header, hash, and every Exif value you rely on.

⚠ **Rename only the exported copy, never the file in the image.**

LESSON 5.1 · SUMMARY

Lesson Summary

Three ideas that lead to one habit

1

Hexadecimal and headers

Two hex digits per byte; a file's first bytes, such as FF D8, give its real type.

2

Renamed files

Extensions can lie; compare them with headers, as forensic tools do.

3

Photo metadata

Exif holds device, settings, time and place; check the clock and time zone.

Trust the contents, and corroborate the metadata.

UP NEXT

Keyword Searching and Carving

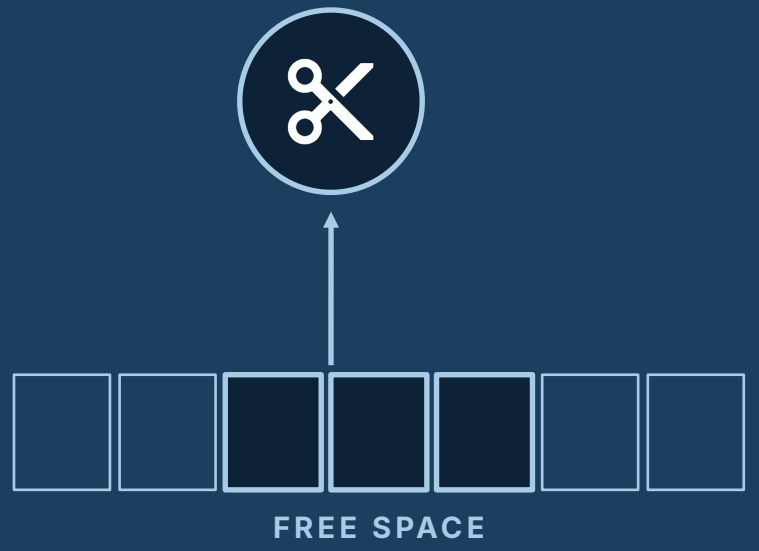
Next, a lab: search the whole image, and recover files from free space.

- **Keyword searching**
- **Carving files**
- **Tagging and reporting**

DFBOOK DIGITAL FORENSICS · LESSON 5.2

Keyword Searching and Carving

Finding words, and whole files, anywhere on a drive



IN THIS LESSON

1. Keyword searching
2. Carving files
3. Tagging and reporting

LESSON 5.2 · OVERVIEW

Keyword Searching and Carving

Evidence is not always in a named file. A keyword search finds words anywhere on the image; carving rebuilds whole files from free space where no record is left.

BY THE END YOU WILL BE ABLE TO

1. **Plan and run a keyword search**
2. **Explain how files are carved from free space**
3. **Tag findings and produce a report**

INTRODUCTION · WHY THIS MATTERS

Why This Matters

Illustrative: photos disguised to slip past an email filter



Photos taken

Inside a new factory

Headers altered

First bytes changed to zzzz

Sent out

The filter saw no images

A search for a fragment of the JPEG label found one file; a second search for zzzz found the rest.

Searching the raw data finds what file names and viewers hide.

INTRODUCTION · ROADMAP

Your Route Through Lesson 5.2

Each part answers one question, in order

1

PART 1

Keyword searching

How do you search a whole image well?

2

PART 2

Carving files

How are files rebuilt from free space?

3

PART 3

Tagging and reporting

How do you keep track of what you find?

One case runs through the module: Teratai Supplies, where auditors have found payments for materials that were never delivered.

PART 1 OF 3

Keyword Searching

Finding words and patterns anywhere in an image

01

IN THIS PART YOU WILL LEARN TO

- Explain how forensic searches work
- Build and refine a keyword list

KEY TERMS

Index **Hit** **False positive**

PART 1 · CONCEPT

How a Forensic Search Works

It searches the data itself, not just file names

Index first

The tool indexes the text in the image, so later searches return at once.

Exact or substring

Exact match finds whole words; substring finds the text inside longer words.

Two encodings

Windows often stores text in Unicode; make sure both forms are searched.

Everywhere

Files, deleted files, slack and unallocated space.

False positive: a search hit that is not what you are looking for. Every hit must be checked before it counts.

PART 1 · METHOD

Building a Keyword List

Drawn from the case, written down before you search

Names

People, companies and suppliers

Numbers

Invoice and purchase order numbers

Addresses

Email addresses and website names

Phrases

Distinctive words from documents

1. **Refine as you go** — Too many hits? Use exact match or longer terms. None? Try substring, or other spellings.
2. **Keep a record** — List every term, its settings and the number of hits in your notes.

PART 1 · WORKED EXAMPLE

Searching for Altered Photos

Illustrative: the disguised factory photos on a USB drive

Search	Setting	What happened
`FIF`	Exact match	No hits: the label sits inside longer text
`fif`	Substring	Hits, including one file whose header shows zFIF
`zzzz`	Substring	Every file whose first bytes were replaced

Each hit was checked in the Hex view: the first bytes FF D8 FF E0 had been overwritten with z.

ANSWER

The photos were disguised on purpose. Tag them, export copies, and restore the header on a copy to view each one.

PART 2 OF 3

Carving Files

Rebuilding files from free space, with no record left

02

IN THIS PART YOU WILL LEARN TO

- Explain how carving works
- Know its limits

KEY TERMS

Carving **Header** **Footer**

PART 2 · CONCEPT

What Carving Is

Finding a file by its own markers, not by the file system



Carving: recovering a file, or fragments of one, from unallocated or slack space using known patterns, such as a header and footer. Also called salvaging.

When to use it

When a deleted file's record has gone, so a normal recovery finds nothing.

What it needs

Knowledge of file headers, footers and structures, like those in Lesson 5.1.

PART 2 · SEQUENCE

How Tools Carve

Automatic carving, and what can still go wrong



Scan for headers

Across free space and slack

Find the end

A footer, or a size in the header

Save as a file

A new file, with a new name

Check it

Open it, and hash it

LIMITS

Fragmented files

Pieces in different places may not join up.

Overwritten parts

New data may have replaced part of the file.

Damaged headers

Repair from a known good header of the same type.

PART 2 · WORKED EXAMPLE

Carving One Photo by Hand

Illustrative: offsets found in free space with a hex editor

[ ] WHAT THE EDITOR SHOWS

Header FF D8 FF E0

at offset 0x1A000

Footer FF D9

at offset 0x1F3C1

Last byte of the file

0x1F3C2 (the D9)

1 **Work out the length**

0x1F3C3 - 0x1A000 = 0x53C3 = 21,443 bytes

2 **Copy those bytes**

Select from the header to the footer, into a new file

3 **Save and check**

Name it .jpg, open it, hash it

ANSWER

A 21,443-byte JPEG. Record the start and end offsets, so anyone can repeat the carve.

PART 3 OF 3

Tagging and Reporting

Keeping track of every finding, then the lab

03

IN THIS PART YOU WILL LEARN TO

- Tag and extract findings
- Search and carve the training image

KEY TERMS

Tag **Follow up** **Report**

PART 3 · SEQUENCE

Tag, Extract, Report

How Autopsy and similar tools keep findings organised

Tag

Mark each relevant item, for example as Follow Up.

Extract

Save copies of tagged files to your export folder.

Report

Generate a report of the tagged results, as a spreadsheet or web page.

Tags let you return to a finding, share it with a colleague, and show exactly what your report is based on.

Update your examiner notes as you tag, not afterwards.

PART 3 · LAB

Searching and Carving the Image

Case thread: the supplier's name, and deleted images

1. **Check the image**
   Confirm its hash before you start.
2. **Search the name**
   The supplier's name: exact, then substring if needed.
3. **Check every hit**
   Open each one; tag the relevant ones as Follow Up.
4. **Carve images**
   Run Autopsy's carving module over free space.
5. **Review the carved files**
   Open, hash and tag any that matter to the case.
6. **Report**
   Generate a report of the tagged items; update your notes.

⚠ **If a carved image could be illegal, stop at once and tell your instructor. Do not open, copy or keep it.**

LESSON 5.2 · SUMMARY

Lesson Summary

Three ideas that lead to one habit

1

Keyword searching

Index, then search everywhere; exact or substring; both encodings; check every hit.

2

Carving files

Header to footer, from free space and slack; fragments and damage limit it.

3

Tagging and reporting

Tag, extract and report, recording every search and offset.

Every hit is a lead until you have checked it.

UP NEXT

How Data Is Hidden

Next: common ways suspects hide data, and how examiners detect each one.

- **Simple hiding**
- **Encryption and passwords**
- **Steganography**

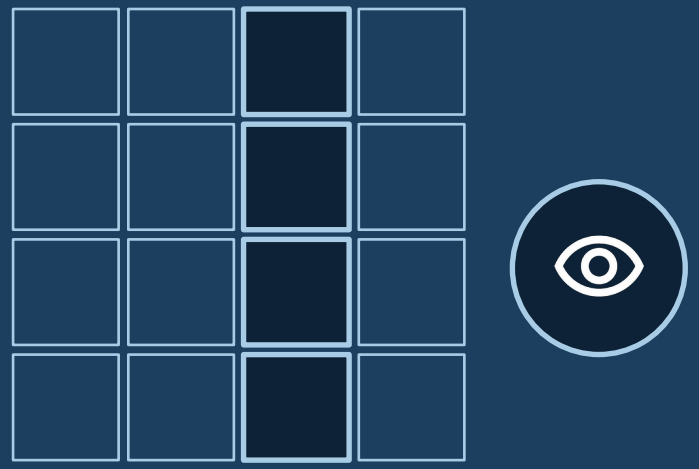
DFBOOK DIGITAL FORENSICS · LESSON 5.3

How Data Is Hidden

Common hiding techniques, and how each one is found

IN THIS LESSON

1. Simple hiding
2. Encryption and passwords
3. Steganography



HIDDEN IN PLAIN SIGHT

LESSON 5.3 · OVERVIEW

How Data Is Hidden

Suspects hide data in many ways, from renaming a file to concealing a message inside a photo. Every technique leaves clues, if you know what to look for.

BY THE END YOU WILL BE ABLE TO

1. **Describe simple hiding techniques**
2. **Explain encryption and password recovery**
3. **Explain how steganography works and is detected**

INTRODUCTION · WHY THIS MATTERS

Why This Matters

Case study from history: a message hidden under a messenger's hair



Head shaved

A message is tattooed on the scalp

Hair grows back

The message is now hidden

Messenger travels

Shaved again on arrival to read it

Rulers in ancient Greece used this to send secret orders. It worked, until their enemies learned to look.

Hidden data is only hidden from someone who does not know where to look.

INTRODUCTION · ROADMAP

Your Route Through Lesson 5.3

Each part answers one question, in order

1

PART 1

Simple hiding

What quick tricks hide files, and how are they caught?

2

PART 2

Encryption and passwords

What can you do with locked data?

3

PART 3

Steganography

How is a message hidden inside a photo?

One case runs through the module: Teratai Supplies, where auditors have found payments for materials that were never delivered.

PART 1 OF 3

Simple Hiding

Quick tricks, and the checks that catch them

01

IN THIS PART YOU WILL LEARN TO

- Name simple hiding techniques
- Match each one to how it is found

KEY TERMS

Hidden attribute **Bit-shifting**

PART 1 · COMPARISON

Simple Hiding Techniques

Each technique, and how examiners detect it

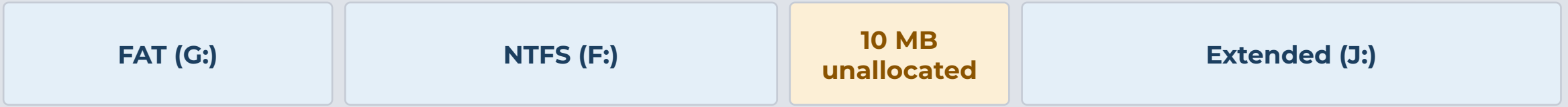
Technique	How it is detected
Changed extension	Compare each file's header with its extension
Hidden attribute	Show hidden files; forensic tools list them anyway
Hidden partition	Account for all disk space; tools show every partition
Clusters marked bad	Check clusters marked bad in a FAT table for data
Bit-shifting	Shift the bits back in a hex editor, and see if data appears

Bit-shifting: moving the bits in each byte left or right, so readable data looks like program code.

PART 1 · WORKED EXAMPLE

A Gap Between Partitions

Illustrative: Windows Disk Management view of an evidence drive



1 What is a normal gap?

On recent Windows, small: about 1 MB before an extended partition.

2 What stands out?

A 10 MB gap with no drive letter, larger than normal.

3 How do you check it?

Open the image in Autopsy, which lists unallocated areas with their sectors.

ANSWER

The 10 MB gap may hold a hidden or deleted partition. Examine it, and record its start and end sectors.

PART 2 OF 3

Encryption and Passwords

Locked files, and the lawful ways to open them

02

IN THIS PART YOU WILL LEARN TO

- Explain the problem encryption poses
- Describe password recovery methods

KEY TERMS

Key escrow **Brute force** **Salting**

PART 2 · CONCEPT

Encrypted Files

Unreadable without the password, and very hard to break

Two kinds of program

Encryption tools, and archive tools such as 7-Zip or WinRAR with a password.

Key escrow

Some products keep a recovery key, in case a user forgets the password.

Breaking it

Strong encryption can take years to break with today's tools.

Asking for it

Often the best route is for the suspect to reveal the password.

In Malaysia, Criminal Procedure Code section 116B covers access to computerised data in a search, including passwords and decryption. Confirm with a legal adviser.

PART 2 · COMPARISON

Password Recovery Methods

How forensic password tools work, from slowest to fastest

Method	How it works
Brute force	Tries every combination; always works in the end, but can take far too long
Dictionary	Tries words from dictionaries, in many languages
Hybrid	Adds words from the suspect's life: family, pets, schools
Rainbow table	Looks up pre-calculated hash values, so it is much faster

Salting: adding extra characters to a password before hashing it. It defeats rainbow tables, so cracking takes longer.

PART 2 · WORKED EXAMPLE

A Password-Protected Archive

Illustrative: a locked ZIP file found in a user's documents

WHAT IS KNOWN

The archive is password-protected. The user's other files mention a pet's name, a school and a car.

QUESTION

What is the best first approach?

1. **Check the legal route**
   Can the password be required from the user?
2. **Build a word list**
   Names and words found in the evidence itself.
3. **Try a hybrid attack**
   The word list plus common changes, before brute force.

ANSWER

Ask lawfully first; then a word list from the evidence. Record every attempt and tool used.

PART 3 OF 3

Steganography

Hiding a message inside another file

03

IN THIS PART YOU WILL LEARN TO

- Explain insertion and substitution
- Describe how steganography is detected

KEY TERMS

Steganography **Steganalysis**

PART 3 · CONTRAST

Two Ways to Hide a Message

Steganography: from the Greek for hidden writing

INSERTION

Adding hidden data

- Puts secret data inside a host file
- Not shown when the file is viewed normally
- Example: hidden text in a web page's code

SUBSTITUTION

Replacing bits

- Replaces the least important bits of a host file
- The change is too small for the eye to see
- Example: a message in a bitmap image

Steganalysis: detecting and analysing steganography. To find inserted data, compare what a file shows with what it contains.

PART 3 · WORKED EXAMPLE

Hiding One Letter in Four Pixels

Illustrative: the letter A (01000001) hidden two bits at a time

Pixel	Original byte	Bits of A	Altered byte
1	11101100	01	**11101101**
2	11101100	00	11101100
3	11101100	00	11101100
4	11101100	01	**11101101**

ANSWER

Only the last two bits of each byte change, so the colour barely shifts. Four pixels carry one letter.

PART 3 · METHOD

Detecting Steganography

Clues to look for, and why the original file matters

Duplicate images

Two copies of a picture with different hash values

Stego programs

Steganography tools installed on the drive

Odd changes

File size, image quality or extension changed

Hardest Only the suspect file is available (stego-only).

Easier The original and the altered file can be compared (known cover).

PART 3 · DEMO

Hidden Text and a Stego Image

Demo: two examples of steganography, and how each is found

A web page

1. Viewed in a browser, it shows ordinary text
2. The page's source code reveals hidden text
3. Compare what it shows with what it holds

A photo and its original

1. The two look the same to the eye
2. Their hash values differ
3. Steganalysis compares them to find the message

If steganography is suspected, look for the original file: it is the easiest way in.

LESSON 5.3 · SUMMARY

Lesson Summary

Three ideas that lead to one habit

1

Simple hiding

Extensions, hidden attributes, partitions, bad clusters and bit-shifting, each with a check.

2

Encryption and passwords

Lawful requests first; then dictionary, hybrid, brute force or rainbow tables.

3

Steganography

Insertion or substitution; look for duplicates, stego tools and the original file.

Account for every byte: hidden data leaves a trace.

UP NEXT

Filtering Known Files and Building a Timeline

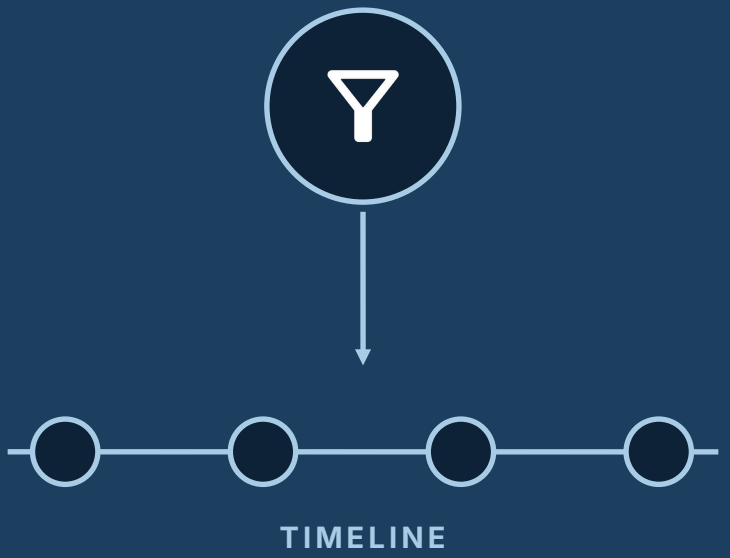
Next, a lab: set known files aside, put events in order, and plan the examination.

- **Filtering known files**
- **Building a timeline**
- **The examination plan**

DFBOOK DIGITAL FORENSICS · LESSON 5.4

Filtering Known Files and Building a Timeline

Set aside what is ordinary, then put events in order



IN THIS LESSON

1. Filtering known files
2. Building a timeline
3. Planning the examination

LESSON 5.4 · OVERVIEW

Filtering and Timelines

A laptop holds hundreds of thousands of files, most of them ordinary. Filtering sets those aside; a timeline shows what happened, and in what order; a plan keeps the work focused.

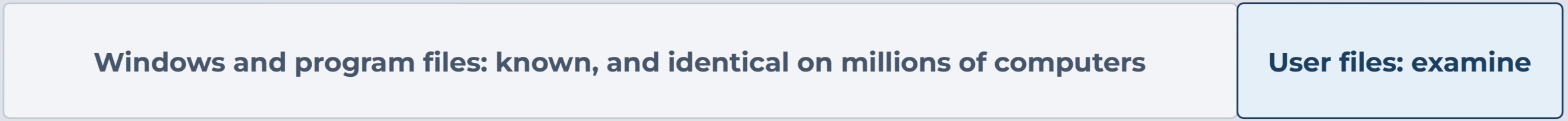
BY THE END YOU WILL BE ABLE TO

1. **Filter out known files with a hash set**
2. **Build a simple timeline from timestamps**
3. **Write an examination plan**

INTRODUCTION · WHY THIS MATTERS

Why This Matters

Illustrative: where an examiner's time goes on one laptop



ALL FILES ON THE DRIVE (NOT TO SCALE)

Filter

Set aside files whose hashes match known software.

Order

Put the remaining evidence on one timeline.

Less noise means more time on the files that can answer the case.

INTRODUCTION · ROADMAP

Your Route Through Lesson 5.4

Each part answers one question, in order

1

PART 1

Filtering known files

How do hash sets cut the work down?

2

PART 2

Building a timeline

How do you put events in a reliable order?

3

PART 3

Planning the examination

How do you keep the work focused?

One case runs through the module: Teratai Supplies, where auditors have found payments for materials that were never delivered.

PART 1 OF 3

Filtering Known Files

Using hash sets to separate ordinary files from the rest

01

IN THIS PART YOU WILL LEARN TO

- Explain known-good and known-bad hash sets
- Filter files in a forensic tool

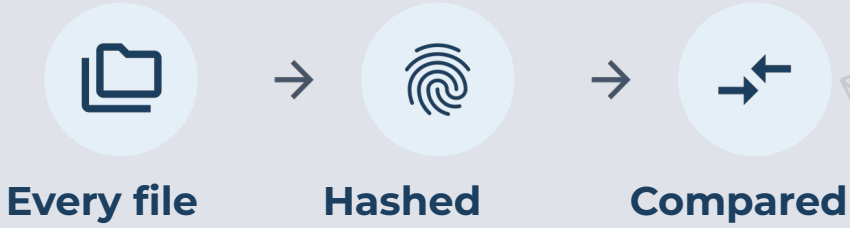
KEY TERMS

Hash set **NSRL** **Known file**

PART 1 · SEQUENCE

How Hash Filtering Works

Every file is hashed, then compared with lists of known hashes



KNOWN GOOD
Windows and program files, such as winword.exe
Set aside

KNOWN BAD
Malware, or illegal material on police lists
Flag at once

UNKNOWN
Everything else, including the user's own files
Examine

Hash set: a list of hash values for known files, used to recognise them whatever their names.

PART 1 · CATEGORIES

Sources of Hash Sets

Where known hashes come from

NSRL

NIST's National Software Reference Library: MD5 and SHA-1 hashes of known software. It holds no illegal files.

Tool libraries

Some suites ship their own, such as FTK's Known File Filter, which also lists known bad files.

Your own

Hash a fresh installation of the systems and programs your organisation uses.

Block-wise hashing: hashing a file sector by sector, so fragments of a known file can be found even after it is deleted.

PART 1 · WORKED EXAMPLE

Filtering One Folder

Illustrative: 1,000 files compared with the NSRL

Result	Files	What to do
Known good	940	Set aside, but keep them in the case
Known bad	0	Nothing to flag
Unknown	60	**Examine every one**

One of the 60 is named winword.exe, but its hash is not in the NSRL.

ANSWER

Examine 60 files, not 1,000. A known name with an unknown hash deserves a close look.

PART 2 OF 3

Building a Timeline

Putting events from many sources in one reliable order

02

IN THIS PART YOU WILL LEARN TO

- Name the main sources of timestamps
- Convert and order times you can defend

KEY TERMS

Timeline **Time zone** **UTC**

PART 2 · CATEGORIES

Where Timestamps Come from

The sources covered so far in this module

File system

Created, modified and accessed times (Lesson 4.2)

Photo Exif

When a photo was taken, by the device clock (Lesson 5.1)

Recycle Bin

When a file was deleted (Lesson 4.4)

Web history

When each site was visited (Lesson 4.4)

Timeline: a list of events from one or more sources, put in order of date and time, with the source of each.

PART 2 · METHOD

Making Times Comparable

Four checks before you put events in order

1	**Check the computer's clock**	Compare its date and time with the real time when it is seized; record any difference.
2	**Convert to one time zone**	Some sources record UTC; Malaysia is UTC+8.
3	**Know what each time means**	A copied file can show a created time later than its modified time.
4	**Record every source**	Next to each event, note where its time came from.

PART 2 · WORKED EXAMPLE

Four Events in Order

Illustrative: times from four sources on 3 March, clock checked and correct

Event	Source	As recorded	Local time
Invoice file created	File system	09:05	09:05
Photo of empty yard	Exif (UTC)	01:15 UTC	09:15
Supplier site visited	Web history	09:20	09:20
Invoice file deleted	Recycle Bin $I	18:42	18:42

ANSWER

Created, photographed, site visited, deleted. Converting the UTC time puts the photo second, not first.

PART 3 OF 3

Planning the Examination

An objective, a scope and three stages

03

IN THIS PART YOU WILL LEARN TO

- Write an examination objective
- Follow the three stages of an examination

KEY TERMS

Objective **Scope creep**

PART 3 · CONCEPT

Objective and Scope

Broad enough to find the evidence; narrow enough to finish

Preserve first

Secure and image the evidence before any analysis, especially SSDs and flash drives.

Write the objective

What to examine, and what evidence would support or refute the complaint.

Respect the scope

A warrant or court order may limit what you may look for.

Watch for scope creep

If you are asked to look further, record who asked, and why.

Scope creep: an examination growing beyond its original description because of new evidence or new requests.

PART 3 · SEQUENCE

Three Stages of an Examination

Prepare, conduct and report, repeating as new evidence appears

1 · Prepare

- Objectives and keywords
- A hypothesis of what happened
- All possible evidence sources
- Images made; tools chosen

2 · Conduct

- Load the image; index and hash
- Survey the file structure
- Search, tag and extract
- Review: is more evidence needed?

3 · Report

- Analyse what was extracted
- Summarise the findings
- Write the report for approval
- Secure the evidence and records

PART 3 · WORKED EXAMPLE

Writing an Examination Objective

Illustrative: the Teratai USB drive

WEAK

Check the USB drive for anything suspicious.

STRONG

Examine USB drive (make, model, serial as recorded on the custody form) and recover invoices from the named supplier dated January to March 2026, if present.

1. **Names the item**

   Identified exactly, as on the custody form.

2. **Says what to find**

   Specific files, a supplier and dates.

3. **Can be answered**

   Present or not present, with evidence.

PART 3 · LAB

Filtering and a First Timeline

Case thread: a hash filter in Autopsy, then six timestamps in order

1. **Load the hash set**
   Add the NSRL hash set to Autopsy's hash lookup.
2. **Run the lookup**
   Mark known files; review the unknown ones.
3. **Collect six times**
   From the sources your instructor lists, with each source.
4. **Convert and check**
   One time zone; note any clock difference.
5. **Put them in order**
   Build the timeline as a table: time, event, source.
6. **Draft the plan**
   Write an examination objective for the next step.

⚠ **Filtering hides files from view; it never deletes them. Keep everything in the case.**

LESSON 5.4 · SUMMARY

Lesson Summary

Three ideas that lead to one habit

1

Filtering known files

Hash every file; set aside known good, flag known bad, examine the unknown.

2

Building a timeline

File system, Exif, Recycle Bin and web history times, in one zone, with sources.

3

Planning the examination

Preserve, set an objective and scope, then prepare, conduct and report.

Start with a plan; follow the evidence; record why.

UP NEXT

Lesson 5 Summary

Next, a short review brings Lessons 5.1 to 5.4 together.

- **Headers and metadata**
- **Searching, carving, hiding**
- **Filtering and timelines**

DFBOOK DIGITAL FORENSICS · LESSON 5 · SUMMARY

Examining Files: Lesson Review

From stored data to findings and a timeline, in one short review



IN THIS LESSON

1. Headers and metadata
2. Searching, carving and hiding
3. Filtering, timelines and plans

LESSON 5 · REVIEW

Lesson 5 at a Glance

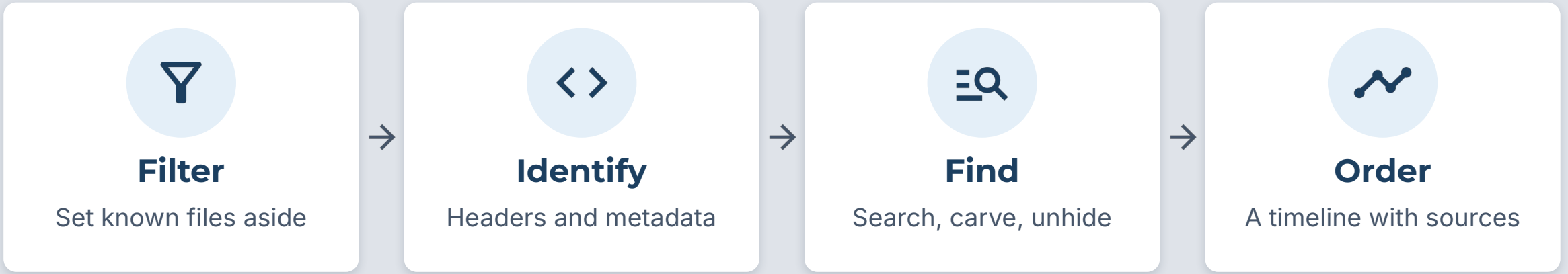
One key point from each sub-lesson

5.1 Signatures and Exif	Headers such as FF D8 reveal a file's true type; Exif needs corroboration.
5.2 Search and carve	Search everywhere and check every hit; carve from header to footer.
5.3 Hidden data	Each hiding technique has a check; the original file unlocks steganography.
5.4 Filter and timeline	Set known files aside; order events in one time zone; plan the work.

LESSON 5 · REVIEW

From Data to Findings

How the four sub-lessons fit together



Trust the contents, check every hit, and record the source of every time.

UP NEXT

Mobile Devices, Email and Social Media

Lesson 6 turns to the sources financial investigators meet most.

- **Mobile devices**
- **Email**
- **Social media**

